One commenter stated a reduction in rebates would lead to the exit of liquidity providers, harming competition,[1490] a lower access fee cap is expected to reduce the fees which are used to fund rebates and consequently rebates are expected to also see a reduction. The Commission acknowledges that profits of liquidity providers may fall because for stocks that remain tick-constrained, the access fee represents a transfer from liquidity demanders to liquidity providers. However, a net decrease in competition could serve to widen the spread beyond a single tick and increase the proceeds of liquidity provision thus incentivizing liquidity provision. While the Commission anticipates and is sensitive to costs to some affected parties, the Commission expects investors, more broadly to benefit.[1491]

Another commenter stated that lowering the access fee cap would result in a "liquidity gap"; that “it is unlikely that the liquidity gap would be met by other market participants;” and that "as spreads further widen, any cost savings non-retail investor participants realize from a reduction in the access fee cap are likely to be more than consumed by the rising frictional costs."[1492] The Commission disagrees that lowering the access fee cap would result in a liquidity gap that market participants would not be able to fill, because if quoted spreads widen beyond any reduction in maker rebates, liquidity providers would stand to earn higher proceeds by supplying at the wider spread. The Commission believes that competition among liquidity providers will keep the cost of accessing liquidity from rising on average in securities where the minimum quoting increment is not a meaningful constraint.[1493] In those stocks that trade with a

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[1490] See Virtu Letter II at 19, Cboe Letter III at 3-4.

[1491] Some Commenters agree. See, e.g., Verret Letter I at 9, Retirement Coalition Letter at 2, Themis Letter at 7, ASA Letter at 4.

[1492] See Cboe Letter IV at 3, 5; see also Cboe Letter III, at 6.

[1493] See section VII.B.3.

quoted spread equal to the tick size, the Commission expects that the reduction in access fees will reduce the net cost of accessing liquidity.[1494] Therefore, non-retail investors would likely see a reduction in overall frictional costs. Finally, one commenter stated that reducing the access fee cap for stocks priced less than $1.00 would impact an exchange's ability to differentiate itself, and the estimated decrease in transactions revenue from these stocks would limit its investments in innovation and technologies.[1495] The impact on an exchange's ability to offer different fees and rebates for stocks priced less than $1.00 is not likely to be large as there is not a substantial degree of differentiation across exchanges currently.[1496] Most exchanges charge fees near or at the fee cap to liquidity takers, and only two exchanges offer rebates to liquidity takers. This is unlikely to change following a decrease in the fee cap. The Commission acknowledges a loss in revenue due to the reduction in rebates for stocks priced below $1.00. It is possible that this could impact exchange investment in new technologies. However, as discussed above in this section, the amendments to Rule 612 are anticipated to lead to more volume on exchange, and hence more trading revenue to exchanges, offsetting this effect. Moreover, as discussed earlier in this subsection, it is necessary to conform rebates for stocks priced below $1.00 with those for stocks priced above $1.00.

One commenter suggested implementing the amendments to reduce the access fee caps before the minimum pricing increments to isolate the impact of the access fee cap on its own.[1497] The Commission has separately considered the impact of the amendments to Rule 612 with the

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1494 Id.

1495 See Cboe Letter II at 9; see also Cboe Letter IV at 1.

1496 See supra Table 4 and surrounding discussion.

1497 See State Street Letter at 5, stating: “We recommend … [i]mplementing any changes to access fee caps before changing quoting increments, to isolate and evaluate the effects. This includes examining whether reducing the access fee cap may affect a security’s designation as ‘tick-constrained.’”

new access fee in place; specifically, the change in the access fee will cause quoted spreads to widen, which may cause some stocks that currently would qualify for a reduction in their tick size under the adopted amendments to Rule 612 to no longer qualify for such a reduction.[1498] However, many stocks will continue to qualify. While the Commission acknowledges that postponing amendments to Rule 612 would allow for time to study the access fee cap in isolation, there is no mechanism by which a reduction in the access fee cap, on its own, would yield the full benefits of the proposed amendments to Rule 612.

The Commission acknowledges that changing the access fee cap at the same time that the changes to Rule 612 are implemented will cause some stocks assigned to a narrower tick size to immediately trade with a quoted spread too wide to qualify for continued assignment to the $0.005 tick size bucket. However, a delay in implementing changes to Rule 612 would delay the accrual of the other benefits the Commission has identified for these changes.

d. Other Effects of the Access Fee Cap Reduction

The Commission anticipates additional benefits inherent to adopting a lower access fee cap for all securities.

Access fees that fund rebates contribute to complexity in markets because they separate both the true cost of demanding liquidity and the proceeds from supplying liquidity, as represented by the quoted half-spread. Commenters stated that lowering the access fee cap to 10 mils would reduce complexity; one commenter stated in the context of supporting a significant reduction in exchange access fees, that current pricing models "contribute to market complexity by encouraging rebate arbitrage strategies and the proliferation of new order types and trading

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[1498] See supra Table 7, note a.

venues designed to exploit different transaction pricing models."[1499] Similarly, a second commenter supported a 10 mil fee cap and stated that maker-taker models, "introduce unnecessary market complexity through proliferation of new exchange order types (and new exchanges) designed solely to take advantage of pricing models."[1500] The same commenter stated that maker-taker pricing may drive orders off exchanges to avoid access fees, and "benefit sophisticated market participants, like market makers and proprietary traders, at the expense of other market participants."[1501]

One manifestation of this complexity is the potential conflict of interest between broker-dealers and their customers.[1502] Multiple commenters stated that a benefit of a lower access fee cap is that it would mitigate such potential conflicts of interests.[1503] Other commenters disagreed.[1504] Some commenters state that due to the complexity, opacity, and potential conflicts inherent in the rebate structure, the Commission should go further than in the current adopted amendments and ban rebates altogether. [1505] The Commission agrees that lowering the access fee cap reduces complexity and may help alleviate potential conflicts of interest.

Finally, the reduction in the access fee cap will improve market quality for stocks that remain tick-constrained. While the amendments to Rule 612 create a smaller tick size for stocks with narrow spreads, spreads naturally vary over time and this variation introduces the possibility

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1499 See Vanguard Letter at 6.

1500 See BMO Letter at 3.

1501 See id.

1502 See supra note 1518 and surrounding text discussing the potential conflicts of interests that exchange fees and rebates may introduce.

1503 See infra note 1514.

1504 See, e.g., Nasdaq Letter I at 2.

1505 See Harris Letter at 4-5, and We The Investors Letter I at 7, arguing that the Commission should go further and ban the use of rebates.

that some stocks could be misassigned to a tick size because trading in the operative period differs from the evaluation period due to factors exogenous to the tick change.[1506] Panel A of Table 10 shows that approximately 13.7% of aggregate share volume will be a false negative under the amendments[1507]—that is, this volume will be assigned a penny tick, but will trade at a TWAQS below $0.015 and therefore trade with a tick-constrained spread a majority of the time. Moreover, some stocks may remain tick-constrained, even at the new half a penny tick. Reducing the access fee cap would lower the cost of accessing liquidity because fee-funded rebates serve as a pure tax on liquidity demanders whenever the spread is tick-constrained (creating a wealth transfer from liquidity demanders to liquidity providers); the reduction in the access fee cap will also reduce the excess supply of liquidity at this price floor.[1508]

3. Exchange Fees and Rebates Determinable at the Time of Execution

In the current environment, as discussed in section VII.C.2, market participants often have to make trading decisions without the ability to determine the exchange fees and rebates they incur at the time of execution, and a market participant's total cost of trading can vary by a significant amount for orders with the same quoted execution price once exchange fees and rebates are accounted for. Current exchange fees and rebates are often based on the participant's relative contribution to the exchange's monthly trading volume during the contemporaneous month, and market participants need to grapple with the uncertainty of forecasting future market

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1506 See section VII.D.1.d for discussion and analysis of the tradeoffs inherent in tick assignment.

1507 This number corresponds to an evaluation period of three months and an operative period of six months, which are the parameters of the rule text. See section VII.D.1.d.

1508 As discussed above (e.g., supra section VII.B.2), a binding price floor on liquidity results in more liquidity supply than demand. Maker-taker pricing exacerbates this problem by taxing demand and subsidizing supply at the price floor. By reducing the access fee, the excess supply is lessened. The lower cost of accessing liquidity can also extend to stocks which trade with a TWAQS greater than $0.015 to the extent to which these stocks may occasionally trade with a spread equal to the tick size.

outcomes should they wish to know what their trading costs are at the time that they execute a trade.[1509] Requiring fees and rebates to be determinable at the time of execution will result in the benefits of increased transparency as to what fees and rebates broker-dealer members are committed to pay when they trade, reducing potential conflicts of interest, and potentially improving broker-dealer routing decisions. These amendments will also result in costs to exchanges associated with revising existing fee schedules to bring them into compliance with the adopted amendments.

The Commission received comments from a broad range of commenters who expressed support for Proposed Rule 610(d) because it would provide enhanced transparency surrounding transaction fees and rebates and alleviate concerns related to potential conflicts of interest.[1510] For example, one commenter stated that it agreed with the Commission's analysis of the benefits of making fees and rebates determinable at time of execution.[1511] Another commenter stated that it agreed with the Commission's assessment "…of how existing exchange pricing tier models can negatively impact market participants behavior."[1512] A third commenter stated that the Rule will "help to make overall trading costs more transparent."[1513]

Multiple commenters pointed out the potential for exchanges' pricing models to create a conflict of interest for broker-dealers who route multiple customers' orders to the exchanges.[1514]

1509 See also Proposing Release, supra note 11, at 80292 for a discussion of the complexity of fee schedules and the difficulty in forecasting fees for a contemporaneous period.

1510 See section IV.E for a discussion of the comment file.

1511 See Council of Institutional Investors Letter at 4.

1512 See BMO Letter at 4,

1513 See Retirement Coalition Letter at 2.

1514 See Vanguard Letter at 6 ("These pricing models can create conflicts of interest with a broker's obligation to obtain best execution for a customer . . . "); STA Letter at 7 ("Today, the primary concerns on access fees are how they contribute to the maker/taker or taker/maker pricing models offered by exchanges and the

One commenter stated that the benefits of pricing determinable at time of execution include "help[ing] broker-dealers make better order routing decisions," and reducing order routing incentives based "on achieving a threshold to gain a specific fee or rebate."[1515] Another commenter also stated that exchanges have little incentive to make fees and rebates determinable at the time of trade because the fee and rebate structure creates "captive customers" that direct order flow to a given exchange in hopes of receiving a given fee or rebate tier in a given month.[1516] One commenter agreed that fees determinable at time of execution has "the potential" to facilitate pass-through of fees and rebates to broker-dealers' customers, and thereby alleviate concerns about "perceived" conflicts-of-interest, but characterized such concerns about conflicts-of-interest as "misplaced."[1517]

As discussed in the Proposing Release, access fees create potential conflicts of interest between brokers and end customers to the extent that brokers can route orders to exchanges with worse execution quality for end customers but more advantageous fees (i.e., a low fee or a high rebate) for the brokers, which the brokers do not pass on to end customers.[1518] For example, a broker may route a customer's limit order to an exchange with a high rebate for liquidity provision, but a relatively low fill rate. The end result would be a high rebate payment for the broker but potentially poor execution quality for the customer.

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offshoots of conflicts of interests in the routing of customer order flow by broker dealers."); Retirement Coalition Letter at 2 "the use of rebates creates conflicts of interest, because when an institutional order is sent as a displayed order, the potential for a rebate may influence where a broker sends the order, even when the investor could receive a better execution on another market."); CII Letter at 3 ("The existing system disadvantages institutional investors because we believe rebates create the kinds of conflicts of interest identified in our policy.").

1515 BMO Letter at 4.

1516 See BMO Letter at 4.

1517 See Nasdaq Letter I at 32.

1518 See Proposing Release, supra note 11, at 80330.

One commenter stated that the supposition that rebates present conflicts of interest is not supported with evidence.[1519] There are, however, significant reasons to believe that rebates present a conflict of interest to agency brokers, even if there is uncertainty regarding to what degree those potential conflicts of interest are being acted upon. Namely, as described above, the quality of the execution accrues to the customer while the rebate accrues to the broker, which leads to a clear divergence of interests whenever the best rebate and the highest quality execution opportunities differ.

Having fees and rebates determinable at the time of execution will mitigate these potential conflicts of interest by increasing broker-dealer accountability to their customers.[1520] This is because the broker-dealer will be able to identify which fees and rebates are associated with which customer order, which at present is not possible at the time of execution.[1521] Having information about the fees and rebates paid as the order is filled will also improve a customer's ability to negotiate routing behavior and monitor the effects that fees and rebates have on its broker's order routing decisions and execution quality.[1522]

In addition, fees and rebates being determinable at the time of execution can make it easier for broker-dealers to pass the actual fees and rebates on to the end customer.[1523] Currently,

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1519 See Nasdaq Letter I at 2.

1520 See Proposing Release, supra note 11, at 80330.

1521 See supra note 1092 and surrounding discussion on information that customers can request from broker-dealers on net transaction fees and rebates through Rule 606(b)(3).

1522 One commenter agreed with the Commission's statement in the Proposing Release that fees being determinable only at the end of the month, as they are currently, impedes investors' ability to evaluate best execution and order routing. Council of Institutional Investors Letter at 4. The Commission believes that making fees determinable at time of execution will help investors make these evaluations, which can contribute to these discussions of fees with their broker-dealers.

1523 The Proposing Release discussed how the inability to know the fee or rebate at the time of a trade could render it difficult for a broker-dealer to pass on fees and rebates to customers to help avoid a potential it can be difficult for a broker-dealer to pass on fees and rebates to individual customers because the exchange fee and rebate pricing tier into which a broker-dealer falls, which ultimately determines fees and rebates on an individual trade, is typically based on the broker-dealer's relative activity across the concurrent month and not an individual trade.[1524] With fees and rebates known at the time of execution, it could be possible for a broker-dealer to more quickly and easily determine the amount to be passed back to the customer. To the extent the amendments increase the proportion of exchange fees and rebates that broker-dealers pass through to end customers,[1525] this will benefit investors and also will reduce the potential benefits broker-dealers may receive from routing customer orders to exchanges with lower fees or higher rebates and thereby reduce distortions in customer order execution quality that this may cause.

A commenter agreed with the Commission's assessment in the Proposing Release that the ability of institutional investors and other market participants to evaluate order execution and routing is significantly impeded by a lack of determinability.[1526] A lack of determinability reduces the amount of information that a market participant can use when evaluating order execution and routing decisions. Without the fees and rebates being determinable, broker-dealers

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conflict of interest. See Proposing Release, supra note 11, at 80329. See also Council of Institutional Investors Letter at 4-5.

1524 See supra note 1084 and surrounding discussion on the current practice of volume-based fee tiers. While the Commission has described the tiered structure of many exchange fee schedules, the benefits of fees and rebates being determinable at time of execution do not depend on, or result from, the fee schedules using volume tiers. The amendments do not ban volume tiers; exchanges can continue to offer volume tiers as long as the tier is based on past—rather than future—volume. Likewise, the benefits of determinability apply even if volume tiers did not exist. For example, some exchanges offer incentives to market makers for frequent quoting at the NBBO—the amendments require that such incentives be based on past quoting at the NBBO so that market participants could determine with certainty their fee at the time of execution.

1525 See supra note 1087 and surrounding discussion on the current practice of broker-dealers passing fees and rebates through to customers.

1526 See Council of Institutional Investors Letter at 4-5.

may have difficulty transmitting information about fees and rebates to customers—the broker-dealer could not commit to a fee or rebate at execution, but would rather need to explain that uncertainty regarding fees and rebates could not be resolved until the end of the month—which may impede competition among broker-dealers.[1527] In contrast, under the adopted amendments, it will be possible for broker-dealers to relay such information about fees and rebates incurred by the broker-dealer to the customer at the time of execution.[1528] This will make the information more usable for customers such as institutional investors, increasing their incentives to ask for such information, as well as increasing the ability of the broker-dealer to transmit the information in a timely manner.

One commenter stated that, "[f]ew brokers route directly to the exchanges…Rather, most brokers pay to route their orders through larger '[direct market access] DMA' brokers to gain the benefit of the large brokers' exchange fee tiers. While the Proposal would simplify life for those few large DMA brokers and proprietary trading firms who closely track where they fall on exchange fee schedules, it wouldn't directly help the referenced Market Participants."[1529] The same commenter stated that most Market Participants, "judging by common practice today," would be unable to account for fees, and the requirements would not improve transparency for off-exchange trading where venues "are not required to charge standard fees, and where fees are often held as competitively sensitive secrets."[1530] The Commission agrees that under common

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1527 See Proposing Release, supra note 11, at 80336. Currently, customers' lack of timely information and certainty about the fees and rebates they incur on the execution of a trade can impact their choice of a broker-dealer for that trade.

1528 For example, under the adopted amendments, it might be possible to include such information in a report to the investor following the execution of their order.

1529 See Pragma Letter at 8.

1530 See id. at 8 ("Even if fees are determinable, it will provide little practical transparency for most Market Participants.").